

MARRYUM ZIA

Physics | Medical Physics | Nuclear Medicine Imaging | Artificial Intelligence

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GitHub | Google Scholar | LinkedIn

Research Profile

Physics researcher with interests in applied physics, medical physics, nuclear medicine imaging, and artificial intelligence. Experienced in computational methods, medical image analysis, deep learning, and real-world clinical imaging data. Interested in interdisciplinary research combining physics, medical imaging, and AI for clinically relevant applications.

Education

University of Agriculture Faisalabad

MPhil Physics

Research focus: AI-based medical image analysis, medical physics, and nuclear medicine imaging

CGPA: 3.89/4.00

Faisalabad, Pakistan

Oct 2024 – Present

University of Agriculture Faisalabad

BS Physics

CGPA: 3.90/4.00

Faisalabad, Pakistan

Sep 2020 – Jun 2024

Research Experience

University of Agriculture Faisalabad / PINUM Cancer Hospital

Researcher – AI-Based Medical Image Analysis

Faisalabad, Pakistan

Jan 2026 – Aug 2026

- Developing a deep learning framework for **multi-class classification of thyroid scintigraphic images** using a dataset of **4,639 clinically labelled images** across five diagnostic categories.
- Processing real-world clinical imaging data from **DICOM/PACS records**, including data extraction, anonymization, quality control, image cleaning, labeling, preprocessing, and dataset preparation.
- Implementing and evaluating transfer-learning architectures including **ConvNeXt, DenseNet, and ResNet** using PyTorch for automated medical image classification.
- Evaluating model performance using sensitivity, specificity, precision, F1-score, ROC-AUC, and temporally separated validation; **ConvNeXt-Tiny achieved approximately 91% accuracy** (95% CI: approximately **87.0–94.2%**) on a temporally separated validation set of **258 images**.
- Applying **Grad-CAM explainability** to visualize model attention and improve the interpretability of AI-generated predictions for nuclear medicine imaging.
- Developing a **Streamlit-based clinical demonstration interface** to assess model usability and reliability and provide a practical framework toward future real-time clinical validation.

Medical Physics Experience

PINUM Cancer Hospital

Medical Physicist Intern – Nuclear Medicine Department

Faisalabad, Pakistan

Sep 2025 – Nov 2025

- Gained hands-on experience in a clinical nuclear medicine environment involving **gamma cameras, SPECT/CT imaging, radiation instrumentation, and radiation safety**.
- Participated in quality control and performance evaluation procedures for gamma camera/SPECT imaging systems and related detector instrumentation.
- Developed practical understanding of scintigraphic imaging, detector operation, image acquisition, and image reconstruction in nuclear medicine.
- Gained experience with the safe handling of radioisotopes and radiopharmaceuticals within clinical nuclear medicine workflows.
- Applied concepts of nuclear physics, radiation detection, imaging physics, and radiation protection to real-world clinical applications.

Publications & Manuscripts

AI-Based Diagnostic Model for Thyroid Scintigraphy

Manuscript submitted for publication.

Selected Research Projects

AI-Based Thyroid Scintigraphy Classification

2025 – Present

- Developed a complete medical imaging pipeline covering clinical data extraction, anonymization, preprocessing, deep learning, evaluation, explainability, and clinical demonstration.

Technical Research Report – Optical Materials and Instrumentation

Jan 2024 – May 2024

University of Agriculture Faisalabad

- Reviewed the development and applications of optical materials, including nonlinear crystals, optical fibers, and optical coatings.
- Investigated applications of optical materials in detector windows, beam optics, and radiation-related instrumentation.

Biometric Attendance System

Mar 2023

Undergraduate Physics Project

- Developed a fingerprint-based attendance system using Arduino Uno, R305 fingerprint sensor, RTC module, LCD, and SD card storage.
- Implemented fingerprint enrollment, verification, time-stamped attendance logging, and real-time system feedback.

Lux Meter Development

Dec 2021 – Jan 2022

Undergraduate Physics Project

- Designed and developed a digital lux meter for measuring illumination intensity using a light sensor and microcontroller-based electronic circuitry.
- Applied principles of electronics, sensors, measurement instrumentation, and experimental physics to develop and test the measurement system.

Technical Skills

Programming & Data Analysis: Python, NumPy, Pandas, Matplotlib, Scikit-learn

Deep Learning & Computer Vision: PyTorch, timm, Convolutional Neural Networks (CNNs), Transfer Learning, OpenCV, Grad-CAM

Medical Imaging & Data: DICOM, pydicom, PACS data handling, medical image preprocessing, image classification, image anonymization

Nuclear Medicine & Medical Physics: SPECT/CT, gamma cameras, scintigraphic imaging, radiation detectors, dose calibrators, radioisotopes, radiopharmaceuticals, radiation safety, imaging system quality control

Software & Research Tools: GitHub, VS Code, Google Colab, PyCharm, Arduino IDE, LaTeX, Microsoft Office

Honors & Academic Achievements

- Merit Scholarship recipient during BS Physics studies.
- Academic high achiever / class topper during undergraduate and postgraduate studies.
- Merit Scholarship recipient during MPhil Physics studies.

Research Interests

Physics | Medical Physics | Nuclear Medicine Imaging | Medical Image Analysis | Artificial Intelligence in Healthcare | Deep Learning | Computer Vision | Explainable AI

English Language Proficiency

IELTS Academic – Overall Band 7.0 Listening: 8.0 | Reading: 7.5 | Speaking: 7.0 | Writing: 6.0